

Grid Strain and the Transmogrification Exercise

Executive Summary

The voice note contains two connected ideas. The first is a systems diagnosis: the electric grid is a layered, historically accumulated network that must balance generation, transmission, distribution, protection, markets, regulation, and end-use demand in real time. The second is an exercise concept: when an adversary, hazard, or disruptive event sends harmful “force” toward a system, do not treat the incoming input only as damage; identify whether any portion can be redirected, absorbed, converted, or used to improve resilience.

The most productive engineering interpretation is not “turn a missile into electricity.” A missile’s chemical, kinetic, thermal, and structural energy is released on a timescale and in a location that make safe recovery practically incompatible with protecting people and infrastructure. The defensible and scientifically useful version is: detect and characterize incoming or cascading disturbances, prevent them from reaching critical assets, isolate damaged sections, preserve essential services, recover useful information or controllable flexibility, and restore the system with the least additional loss.

This concept can become a safe simulation laboratory and tabletop exercise called **Transmogrification: Converting Adversarial Inputs into Beneficial Vectors**. It should test decision quality, system observability, islanding and load prioritization, data integrity, cross-organizational coordination, resource allocation, and recovery—not missile interception techniques or operational cyberattack methods.

The Grid Is Not One Machine

The electric grid is better modeled as a hierarchy of coupled networks than as a single uniform machine. Bulk generation and high-voltage transmission move power over long distances; substations transform voltage and route flows; distribution feeders deliver service to homes, businesses, and critical facilities; protection and control systems detect abnormal conditions; markets and planning institutions shape when and where assets are built.

This layered structure is not necessarily evidence that engineers “designed badly.” Much of the network was built incrementally, under different technologies, rules, ownership structures, load forecasts, and geographic constraints. The resulting system contains legacy equipment, local optimization, bottlenecks, incomplete observability, and interfaces that were never designed for today’s combination of distributed energy resources, extreme weather, electrification, storage, and very large digital loads.

A more precise diagnosis is **path dependence plus coordination failure**: earlier design decisions constrain later choices, while responsibility is divided among utilities, system operators, regulators, equipment manufacturers, municipalities, customers, and governments. The International Energy Agency identifies planning, finance, regulation, supply chains, digitalization, and workforce capacity as linked grid-investment challenges; it also reports that

many advanced renewable projects are waiting for connection because grid expansion has not kept pace.^{[1][2]}

Why Strain Is Increasing

North American reliability assessments identify unusually rapid demand growth from data centers, artificial-intelligence infrastructure, electrified transportation, heat pumps, manufacturing, and other large loads. NERC's 2025 Long-Term Reliability Assessment projects more than 224 GW of aggregated summer peak-demand growth and more than 245 GW of winter peak-demand growth over its ten-year horizon, with new data centers accounting for most of the projected increase.^[^3]

The problem is not simply that the world is “running out of electricity.” Reliability depends on location, timing, weather, ramping capability, transmission constraints, reserve margins, fuel availability, equipment condition, and the ability to control flows. A region can have enough annual energy but still face a shortfall during a narrow peak interval or during a transmission outage.

Grid construction also has long lead times. The IEA reports that transmission projects can take more than a decade because of planning, permitting, land acquisition, procurement, public engagement, and construction. This creates a timing mismatch: generation or large loads can be proposed and built faster than the network needed to connect and serve them.^[^4]

Supply chains amplify the delay. Large transformers, conductors, switchgear, protection equipment, and control hardware require specialized manufacturing and skilled labor. The IEA describes growing pressure on transmission-component prices, lead times, and procurement, emphasizing coordinated planning and long-term purchasing.^[^2]

“Misallocation” Needs Precision

The phrase “misallocation of energy” points toward several different technical problems that should not be collapsed into one category:

- **Conversion losses:** energy is lost as heat in generators, transformers, conductors, power electronics, and end-use devices.
- **Congestion:** a lower-cost or cleaner generator may be unable to serve a customer because a line or transformer is full.
- **Curtailement:** available generation is reduced because the system cannot safely absorb or move it.
- **Poor temporal alignment:** supply and demand occur at different times, requiring storage, flexible demand, reserves, or dispatchable resources.
- **Poor spatial alignment:** generation is far from demand, while transmission capacity is limited.
- **Reliability redundancy:** equipment that appears underused may exist to provide reserve capacity, fault tolerance, or restoration capability.

- **Administrative and market friction:** tariffs, interconnection rules, ownership boundaries, and permitting can prevent technically feasible improvements.
- **End-use inefficiency:** buildings, appliances, motors, lighting, cooling systems, and standby loads can consume more energy than necessary.

The useful question is therefore not “Why is all this energy wasted?” but “At which layer is value being lost, and which intervention addresses that layer without creating a larger reliability or safety problem?” The World Bank maintains country-level indicators for transmission and distribution losses, while the IEA documents grid congestion, connection queues, and curtailment as distinct infrastructure problems.^{[5][6]}

Why Bills Can Rise During Infrastructure Spending

A utility bill is not a direct meter of physical energy scarcity. Depending on the jurisdiction, it can include generation costs, transmission and distribution costs, storm recovery, vegetation management, wildfire mitigation, fuel adjustments, capacity or demand charges, taxes, public programs, and the utility’s regulated return. Infrastructure investment may raise near-term costs while reducing future outage risk or enabling new capacity.

This does not mean every expenditure is efficient. A serious exercise should force participants to distinguish between investments that improve reliability, investments that reduce operating cost, investments required for safety or compliance, investments that enable future load, and investments that mainly shift costs among customers. It should also ask who receives the benefit and who pays.

The AI Grid Concept

An AI supervising the entire grid is a compelling image, but a real deployment should be framed as **decision support with bounded authority**, not as an unconstrained autonomous controller. DOE’s Grid Modernization Initiative describes a modern grid in terms of resilience, reliability, security, affordability, flexibility, and sustainability, and emphasizes tools that measure, analyze, predict, protect, and control the system.^[^7]

AI could contribute to:

- Forecasting demand, weather, renewable output, equipment failure, and restoration time.
- Detecting abnormal patterns across operational technology and customer-facing systems.
- Ranking vulnerable substations, feeders, transformers, and critical loads.
- Recommending switching, islanding, storage, demand response, and restoration sequences.
- Identifying congestion, curtailment, and interconnection bottlenecks.
- Comparing competing investment plans under multiple futures.
- Summarizing uncertainty and showing operators why a recommendation was made.

AI should not be allowed to silently optimize for “maximum energy delivered” at the expense of human safety, equipment limits, voltage and frequency stability, privacy, cybersecurity, or

equitable access. NIST’s smart-grid cybersecurity guidance reflects the need to treat communications, control, identity, resilience, and security as integrated design concerns.^{[8][9]}

A useful architecture has four layers:

Layer	Function	Safe design requirement
Observe	Sensors, telemetry, weather, asset condition, customer and facility status	Data provenance, time synchronization, uncertainty labels
Understand	State estimation, forecasts, anomaly detection, scenario analysis	Human-readable explanations and independent validation
Recommend	Switching, load prioritization, storage, mutual aid, restoration options	Constraint checking and competing alternatives
Act	Operator-approved control or tightly bounded automation	Interlocks, fail-safe behavior, audit logs, manual override

The Missile Idea: What Must Be Reframed

The physical intuition is understandable: an incoming object contains stored energy, and it seems wasteful to let that energy become destruction. In practice, energy is only useful if it can be captured at a manageable rate, in a controlled location, with equipment that survives the capture process. A missile or explosive impact releases energy too quickly and unpredictably, while the infrastructure needed to intercept it, contain it, convert it, and connect the output to a grid would itself be a major hazard.

The exercise should therefore exclude instructions for missile interception, explosive-energy capture, weapon modification, targeting, or counterattack. It can still preserve the creative core by treating the “missile barrage” as an abstract **adversarial input** whose physical details are intentionally withheld. The exercise asks what the system can learn, isolate, protect, reroute, recover, and improve.

The concept becomes scientifically defensible when “transmogrification” means one or more of the following:

- **Information conversion:** turn an attack signal, outage pattern, or anomaly into actionable situational awareness.
- **Spatial conversion:** redirect flows through healthy network paths or island critical facilities.
- **Temporal conversion:** shift flexible demand or store energy for a later peak.
- **Asset conversion:** use distributed energy resources, microgrids, batteries, and generators as resilience assets.
- **Operational conversion:** convert an attack into a restoration opportunity by learning the adversary’s pattern and hardening the exposed path.
- **Institutional conversion:** turn fragmented responsibility into a coordinated response structure.

Exercise Name and Purpose

Exercise title: Transmogrification: Beneficial Vectors in a Stressed Grid

Purpose: Explore how a grid operator, utility, emergency-management organization, public agency, and critical-facility community can convert disruptive inputs into bounded resilience actions while maintaining safety, reliability, cybersecurity, and public trust.

Core question: When harmful force enters a coupled system, what can be safely absorbed, redirected, isolated, learned from, or converted into useful flexibility—and what must simply be blocked or allowed to dissipate?

Scope: A fictional regional distribution-and-transmission system during an extreme-heat event, with high demand, constrained transformers, data-center load, rooftop solar, batteries, a hospital, water infrastructure, residential customers, and incomplete communications.

Out of scope: Weapon design; missile interception tactics; explosives; offensive cyber operations; real utility control commands; unauthorized access to operational technology; and claims that destructive energy can be harvested economically after impact.

Simulation Laboratory Version

The simulation lab should begin with a simplified network model rather than a real utility topology. Nodes can represent generation, substations, feeders, critical facilities, storage, flexible loads, and communication links. Edges can carry attributes such as capacity, latency, failure probability, repair time, and dependency on communications.

Each round represents a time interval. Participants receive a state estimate, forecast, asset condition report, and uncertainty level. They choose actions such as reconfiguration, demand response, battery dispatch, microgrid islanding, public communication, mutual aid, equipment conservation, or postponement of noncritical load. The model returns consequences: served load, unserved critical load, losses, congestion, reserve margin, recovery time, equipment stress, and new information.

A minimal state vector could include:

- Available generation and reserve capacity.
- Demand by customer class and geographic zone.
- Transmission and feeder loading.
- Transformer temperature and failure risk.
- Battery state of charge and dispatch limits.
- Weather severity and forecast confidence.
- Communication availability and data integrity.
- Critical-load priorities.
- Repair crews, spare equipment, and estimated restoration times.

The lab should compare at least four strategies:

Strategy	Description	What it tests
Baseline reactive	Respond only after overloads or failures occur	Cost of delayed awareness
Central optimization	One coordinator allocates resources across the network	Benefits and risks of global visibility
Distributed resilience	Local microgrids and facilities act within predefined limits	Value of autonomy and interoperability
Human-AI partnership	AI forecasts and recommends; authorized operators approve	Explainability, trust, and bounded control

The goal is not to produce a single “optimal” result. It is to reveal trade-offs. A strategy that minimizes immediate unserved energy may increase transformer aging, reduce later reserves, create inequitable service, or make recovery harder.

Tabletop Exercise Version

The tabletop should follow HSEEP principles: establish a purpose, define objectives, develop a scenario and injects, facilitate discussion, evaluate performance, and produce an After-Action Report and Improvement Plan.^{[10][11][12]}

Participants

- Utility operations and distribution control.
- Transmission or balancing authority representative.
- Emergency management and public safety.
- Communications and public information.
- Water, healthcare, telecommunications, and transportation representatives.
- Cybersecurity and information-technology personnel.
- Regulator or policy representative.
- Community or customer advocate.
- AI/model-governance or data-science representative.

Suggested Scenario

A prolonged heat wave produces evening demand near forecast limits. A large digital load is operating continuously. A transmission constraint limits imports into the region. Several transformers are running hot. A communications anomaly causes conflicting telemetry. A wildfire or equipment fault removes one feeder. A neighboring region requests assistance while local critical facilities report fuel and battery constraints.

The scenario may use the metaphor of an incoming “barrage,” but each item should be represented as a safe inject: a forecast shock, equipment failure, corrupted data, misinformation, supply delay, sudden load, communications loss, or weather escalation.

Inject Sequence

Inject	Situation	Discussion prompt	Beneficial vector
1	Heat forecast worsens and demand rises	Which loads and assets matter first?	Forecast uncertainty becomes prioritized preparation
2	Two transformer temperature sensors disagree	Which data can be trusted?	Data conflict becomes a validation and redundancy test
3	A feeder trips near a hospital and water plant	What is isolated, and who has authority?	Failure becomes a controlled islanding and critical-load exercise
4	A large load requests immediate energization	What evidence is required before approval?	Demand growth becomes an interconnection-governance test
5	Battery reserves are limited	Should batteries serve peak load, ride through faults, or preserve black start capability?	Stored energy becomes a temporal-flexibility decision
6	Public claims accuse the utility of wasting power	What can be disclosed without exposing security-sensitive information?	Misinformation becomes a trust and transparency test
7	Spare transformer delivery is delayed	What temporary topology and mutual aid options exist?	Supply-chain friction becomes contingency planning
8	Communications fail for one operating zone	Can local controls maintain safe service?	Loss of central visibility becomes an autonomy test
9	Conditions improve but equipment remains stressed	When does restoration become unsafe?	Recovery becomes a lifecycle and damage-avoidance test

Evaluation Objectives

The exercise can test whether participants can:

1. Identify critical loads and establish transparent prioritization criteria within 15 minutes of a confirmed constraint.
2. Distinguish an energy shortfall from a transmission, distribution, equipment, communications, or data-quality problem.
3. State who may authorize islanding, load reduction, switching, public warnings, and restoration.
4. Use AI or analytical recommendations without treating them as authoritative when data quality is uncertain.
5. Preserve cybersecurity and safety boundaries while sharing enough information for coordination.
6. Convert distributed resources into a coordinated resilience portfolio without exceeding equipment or contractual limits.
7. Communicate trade-offs to the public in plain language.
8. Produce at least three specific, measurable corrective actions after the exercise.

HSEEP's evaluation approach supports linking observations to objectives, capability targets, critical tasks, and corrective actions rather than treating an exercise as a performance contest. [13][14]

Metrics and Scoring

Potential metrics should be defined before play begins:

- Time to detect and classify the disturbance.
- Time to identify affected critical facilities.
- Percentage of critical load served.
- Total unserved energy, separated from ordinary customer load.
- Number and duration of unsafe or unauthorized actions.
- Forecast error and confidence calibration.
- Number of decisions delayed by unclear authority.
- Communications restoration time.
- Battery reserve preserved for later events.
- Transformer or equipment stress exposure.
- Equity impact across customer groups.
- Time to develop and assign corrective actions.

A qualitative scoring scale can supplement numerical metrics: **effective**, **partially effective**, **ineffective**, or **not tested**. Evaluators should record not only what decision was made, but what information was available, who had authority, what assumptions were used, and what prevented a better choice.

What Experts Will Reject

Participants may correctly reject several literal claims in the original concept:

- A destructive missile impact is not a practical energy-storage opportunity.
- Grid power cannot simply be “sunk” into an arbitrary incoming object without an engineered conversion path and severe safety risks.
- An AI cannot observe “everything” on a grid; sensing is incomplete, data ownership is fragmented, and some systems are intentionally isolated.
- Lower transmission latency does not automatically mean lower energy loss; voltage, current, conductor resistance, topology, and power quality matter.
- Redundancy is not always waste. Spare capacity can be essential to reliability and recovery.
- A nationally interconnected grid is not necessarily misdesigned because it contains bottlenecks; some constraints reflect geography, cost, land, regulation, reliability standards, or deliberate security margins.
- A counterattack is not the same as resilience. Converting an adversary's action into advantage should not mean escalating or exploiting a vulnerability.

These objections improve the exercise. They separate a useful systems metaphor from an unsafe or physically implausible engineering proposal.

The Re-engineered Prompt

Design and analyze a safe, non-operational simulation and tabletop exercise called **Transmogrification: Converting Adversarial Inputs into Beneficial Vectors**.

The exercise uses the electric grid as a complex adaptive system under stress. It examines how historically layered infrastructure, demand growth, aging equipment, transmission and distribution bottlenecks, renewable interconnection queues, large digital loads, extreme weather, communications failures, and fragmented governance can create reliability and allocation problems.

The central idea is not to harvest energy from missiles or to design counterattacks. Instead, model incoming harmful inputs abstractly as disturbances—equipment failures, cyber-physical anomalies, extreme heat, misinformation, supply-chain delays, sudden demand, or attacks on infrastructure. For each disturbance, identify what must be blocked, what can be isolated, what can be redirected, what can be converted into information, what can be shifted in time, and what can be recovered through distributed resources such as batteries, microgrids, demand response, and mutual aid. Build two linked products:

1. A simulation laboratory using a fictional grid with generation, transmission, substations, feeders, critical facilities, batteries, flexible loads, communications links, uncertainty, and repair constraints.
2. A HSEEP-aligned tabletop exercise with a situation manual, participant roles, scenario, injects, discussion questions, evaluation guide, hot wash, and After-Action/Improvement Plan.

Test observability, data integrity, forecast uncertainty, authority, safety, cybersecurity, islanding, load prioritization, resource allocation, public communication, equity, recovery, and the risks of over-automating decisions with AI.

Use real, authoritative research on grid strain, construction lead times, transmission and distribution losses, congestion, curtailment, interconnection queues, demand growth, reliability, grid modernization, smart-grid cybersecurity, and exercise design. Do not use real utility topology, real control commands, weapon-construction details, explosive-energy calculations, missile interception tactics, or offensive cyber procedures. Make the results useful for education, resilience planning, and systems thinking.

Recommended Next Steps

1. Preserve the original voice note as a **concept record**, not as an engineering specification.
2. Adopt “transmogrification” as a systems-learning metaphor with explicit safety boundaries.
3. Start with a small fictional feeder model before attempting a regional grid model.
4. Define five measurable objectives before writing scenario details.

5. Run a paper tabletop first, then a spreadsheet or code-based simulation, then compare the results.
6. Invite an electrical engineer, utility professional, emergency manager, cybersecurity practitioner, and community advocate to critique the exercise.
7. Treat every AI recommendation as a hypothesis subject to constraints, uncertainty, human authorization, and audit.
8. End with an improvement plan that names an owner, deadline, resource need, and verification method for every corrective action.

Conclusion

The strongest idea in the voice note is not literal energy transmutation. It is a general resilience principle: harmful inputs should be classified quickly, prevented from propagating, converted into information, redirected where safe, and used to improve the system's next decision. The electric grid is an excellent laboratory because it combines physical flows, cyber systems, markets, institutions, human needs, long construction cycles, and high consequences.

The exercise is useful precisely because it can make participants confront the boundary between optimization and resilience. The best response is not always to deliver the most energy immediately. Sometimes it is to preserve reserve, protect a transformer, island a facility, reject an unsafe load, distrust a sensor, communicate a difficult trade-off, or spend money on a component that appears idle until the day it prevents a cascading failure.

References

1. [1 - Investment in electricity networks, including transmission and distribution infrastructure, needs t...](#)
2. [Building the Future Transmission Grid – Analysis - IEA](#) - The urgency to expand and modernise the electricity transmission network infrastructure around the w...
3. [Long-Term Reliability Assessment, January 2026](#) - This 2025 LTRA assessment period includes projections for 2026–2035; however, some figures and table...
4. [World Energy Outlook 2024](#) - At the moment, for every dollar spent on renewable power, 60 cents are spent on grids and storage.
...
5. [Electric power transmission and distribution losses \(% of ...](#) - # Electric power transmission and distribution losses (% of output)
...
Country
Most Recent Year
Mos...
6. [Renewables 2024](#) - Investment in grid infrastructure is lagging, with more advanced projects waiting to be connected, ...

7. About the Grid Modernization Initiative - The Grid Modernization Initiative (GMI) works across the U.S. Department of Energy (DOE) to create t...

8. Release of NIST Interagency Report 7628 Revision 1, Guidelines for Smart Grid Cybersecurity - <https://www.nist.gov/publications/release-nist-interagency-report-7628-revision-1-guidelines-smart-g...>

9. Guidelines for Smart Grid Cybersecurity - <https://www.nist.gov/publications/guidelines-smart-grid-cybersecurity>

If you have any questions abou...

10. Homeland Security Exercise and Evaluation Program (HSEEP) - Exercise Objectives

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11. HSEEP Policy and Guidance - HSEEP Resources - Preparedness Toolkit - The Homeland Security Exercise and Evaluation Program (HSEEP) provides a set of fundamental principl...

12. Homeland Security Exercise and Evaluation Program - HSEEP provides a set of guiding principles for exercise and evaluation programs, as well as a common...

13. Exercise Evaluation Guides - HSEEP Resources - Preparedness Toolkit - # Exercise Evaluation Guides (EEGs)

EEGs are designed to accomplish several goals:

- Streamline data...

14. Cybersecurity Tabletop Exercise Guide - Resource Library - Program (HSEEP): <https://www.fema.gov/sites/default/files/2020-04/Homeland-Security-Exercise-and-Eva...>